

RED SEAL EXAM PREP

310T Truck & Transport Mechanic

Free 50-Question Mock Exam

Practice for the Red Seal / Certificate of Qualification exam

Pass mark: 70% · Aligned to the published occupational standard topic weightings
<https://redsealquiz.ca> · Edition 2026-07

About This Material

Independent, unofficial study resource. This publication is not affiliated with, endorsed by, or approved by the Canadian Council of Directors of Apprenticeship (CCDA), Employment and Social Development Canada, the Red Seal Program, Skilled Trades Ontario, or any provincial or territorial apprenticeship authority. "Red Seal" is used solely to describe the examination for which this material provides practice.

Originality. All questions and explanations in this publication are original works created by the author. This material references only publicly available occupational standards (Red Seal Occupational Standard) for topic alignment and reproduces no content from any actual examination.

No guarantee of results. This study material is provided for educational and informational purposes only. Success on any certification examination depends on many factors, and no representation is made that using this material will result in passing any exam. Exam content, structure, and weightings are set by certification authorities and may change without notice.

Technical accuracy. Code values and technical specifications cited in explanations reflect commonly referenced Canadian standards at the time of writing. Always defer to the current edition of the applicable code or the manufacturer's documentation in real work.

Refunds. Because this is an instantly delivered digital file, all sales are final once the download link has been delivered. If you hit a technical problem — a corrupted or inaccessible file, or a duplicate purchase — contact us within 14 days and we will repair the file or refund you in full.

Updates. Corrections and additions are published to the same file. Re-download from your Ko-fi library to get the current edition at no extra cost.

© 2026 the author. All rights reserved. For personal study use by the purchaser only — redistribution, resale, or file sharing is prohibited.

Mock Exam — answer all questions, then check the key

Q1. What is the function of the quick-release valve in an air brake system?

- A) Quickly increases brake application pressure for emergency stops
- B) Allows air to bypass the chambers when the parking brake is engaged
- C) Regulates maximum brake application pressure to prevent lockup
- D) Rapidly exhausts chamber air when the brakes are released

Q2. A diesel engine's compression test shows one cylinder at 250 PSI while the others are 430–460 PSI. A wet compression test (with oil added) brings the low cylinder up to 415 PSI. What does this indicate?

- A) Head gasket failure between two cylinders — oil seals the breach during the wet test
- B) A cracked cylinder head — the crack vents cylinder pressure into the water jacket
- C) Wear at the piston rings or cylinder bore — the oil temporarily sealed that contact
- D) A burnt exhaust valve — the valve face no longer seals against its hardened seat

Q3. What is the MOST likely cause of excessive engine blow-by (excessive pressure/smoke from the crankcase breather)?

- A) Worn or broken piston rings
- B) Coolant leaking into the crankcase
- C) Faulty crankcase ventilation filter
- D) Engine oil overfilled

Q4. Under National Safety Code Standard 11, an air-braked truck is rejected at a periodic inspection if the low air pressure warning device has not activated by the time system pressure falls below:

- A) 690 kPa (100 psi)
- B) 414 kPa (60 psi)
- C) 552 kPa (80 psi)
- D) 138 kPa (20 psi)

Q5. What colour of diesel exhaust smoke indicates burning engine oil, and what are the common causes?

- A) Blue-grey smoke — oil is entering the combustion chamber through worn seals or rings
- B) Colourless — engine oil combustion produces no visible smoke under normal conditions
- C) White smoke — engine oil burns white because it is a lighter hydrocarbon than diesel
- D) Black smoke — excess fuel causes oil to burn along with diesel fuel

Q6. A relay valve in an air brake system is located at the rear axle brakes. Its purpose is to:

- A) Speed brake application using air from a local reservoir
- B) Automatically adjust rear brake pressure based on load
- C) Reduce maximum brake pressure to the rear axle
- D) Prevent brake pressure from exceeding governor cut-out pressure

Q7. Trucks, tractors and trailers must be inspected every 24 hours and the driver completes a trip inspection report (some carriers call it a vehicle condition report). Under NSC Standard 13, the carrier must retain that report for a minimum of:

- A) 30 days
- B) 1 year
- C) 6 months
- D) 7 days

Q8. What is the purpose of the "dither" signal in a proportional solenoid control system for an electronically controlled transmission?

- A) Dither is a rapid alternating polarity signal that drives the solenoid in both directions to ensure it cannot stick
- B) A high-frequency, low-amplitude oscillation superimposed on the command signal to prevent solenoid stiction
- C) Dither refers to the controller's delay time between receiving a command and activating the solenoid
- D) Dither is a diagnostic mode that cycles solenoids to test for sticking during pre-operation checks

Q9. Engine coolant (antifreeze) must be maintained at the correct concentration. The recommended freeze protection for most Canadian operating conditions is:

- A) 25% antifreeze / 75% water
- B) 75% antifreeze / 25% water
- C) 50% antifreeze / 50% water
- D) 100% antifreeze for maximum protection

Q10. One of the one-way check valves between the supply (wet) reservoir and a service reservoir on a dual-circuit air brake system fails open. What is the consequence?

- A) That service reservoir can no longer be charged from the supply tank
- B) The two service circuits equalize, which improves braking balance
- C) Nothing - the valve only comes into play during parking brake use
- D) A leak in that service circuit can also drain the supply reservoir

Q11. What is the function of a driveshaft's slip yoke?

- A) To allow the driveshaft to rotate at different speeds than the transmission output when cornering
- B) To absorb torque spikes from the engine and protect the transmission output shaft
- C) To allow quick driveshaft removal for service without disconnecting universal joints
- D) To allow the driveshaft to change length as the suspension moves the axle relative to the frame

Q12. A truck body module controls the exterior lighting with PWM outputs. The headlights dim randomly. A DMM on the headlight feed shows a steady average voltage that does not change when the lights dim. What tool would best diagnose the intermittent signal fault?

- A) DMM set to Hz (frequency) mode
- B) A 12-volt incandescent test light
- C) An oscilloscope on the PWM output
- D) Amp clamp around the headlight wire

Q13. A truck is in for a periodic commercial motor vehicle inspection under National Safety Code Standard 11. Every tire gauges within the recommended pressure and every tread depth is above the reject figure. Which of these remaining findings is a reject?

- A) A radial tire beside a non-radial on the same axle
- B) A steer tire built by a different maker than its mate
- C) Dual-mounted tires whose diameters differ by 8 mm
- D) A retreaded tire fitted to one of the drive axles

Q14. After coupling a tractor to a semi-trailer, which check confirms that the fifth wheel is properly locked to the trailer kingpin?

- A) A tug test plus visual confirmation that the jaws are locked around the kingpin with no coupler gap
- B) Raising the landing gear fully and confirming the trailer follows the tractor when driving forward
- C) Checking that the fifth wheel plate is well greased and the kingpin shows witness marks
- D) A tug test alone — if the tractor cannot pull out from under the trailer, the coupling is complete

Q15. A heavy truck carries four 12-volt batteries wired in parallel. One is replaced with a new battery of the same type while the other three stay in service. What is the usual consequence?

- A) The bank still performs at the level of its weakest battery
- B) The bank now supplies 48 volts because the batteries are in parallel
- C) The alternator will overcharge because bank resistance has dropped
- D) The new battery raises the whole bank back to full cranking output

Q16. A truck brake drum of 420 mm nominal size carries no wear limit marking, and no OEM or industry figure is available. Under National Safety Code Standard 11, the drum is rejected once its measured diameter exceeds the original by more than:

- A) 2.3 mm more than the original drum diameter
- B) 0.5 mm more than the original drum diameter
- C) 3.0 mm more than the original drum diameter
- D) 7.5 mm more than the original drum diameter

Q17. A truck clutch is slipping — engine RPM rises sharply during acceleration but vehicle speed does not increase proportionally. What would cause clutch slippage?

- A) Clutch disc too thick — excess disc material increases friction surface contact
- B) Oil or grease contamination on the clutch disc friction surfaces from a leaking rear main seal
- C) Excessive clutch free travel — too much free play causes the pressure plate to apply too much force
- D) Incorrect flywheel surface roughness — a smooth flywheel increases friction and causes slippage

Q18. A high-mileage highway tractor with a non-synchronized 10-speed transmission pops out of gear into neutral while coasting down grades. The complaint occurs in one gear only. What is the most likely cause?

- A) Worn clutching teeth on that gear and its sliding clutch, tapered by repeated pop-out
- B) Loose transmission mounts letting driveline vibration shake the gear out of engagement
- C) A worn blocker ring in the synchronizer for that gear letting the collar back out
- D) A slipping clutch, which pushes the transmission out of gear when the driver lifts off

Q19. When checking a driveshaft working angle, what is the maximum acceptable operating angle for most heavy truck driveshafts to prevent vibration and U-joint wear?

- A) 12 degrees
- B) 3 degrees
- C) 7 degrees
- D) 1 degree

Q20. A truck engine's fuel consumption has increased by approximately 15% compared to 3 months ago with no change in load or routes. What are the MOST likely causes to investigate?

- A) The fuel filter is clogged — this always increases fuel consumption
- B) The air filter is dirty — less airflow raises consumption proportionally
- C) Leaking injectors, timing drift, boost leaks, or parasitic loads
- D) The engine is fine — fuel consumption naturally increases with age

Q21. On a truck with no published manufacturer wear limits, which steering finding does National Safety Code Standard 11 Part B list as a hazardous condition rather than only a reject?

- A) A split steering gear bellow with its clamp still in place
- B) A ball and socket joint with 4 mm play along the ball shank
- C) Centre to full left is one turn more than centre to full right
- D) A tire clearing the frame by 20 mm at one point in the turn

Q22. During a roadside inspection a truck with 10 service brakes is found with three brakes 6.4 mm (1/4 in) or more beyond the brake adjustment limit. Under the out-of-service criteria used at roadside in Canada, can the vehicle continue operating?

- A) Yes — up to 5 brake defects are allowed before out-of-service applies
- B) The 20% rule applies only to trailers, not tractor brake circuits
- C) No — 20% or more defective brakes puts the vehicle out of service
- D) Yes — only locked-up or missing brakes are cause for out-of-service

Q23. A truck's clutch pedal free travel is too excessive (more than specified). What is the effect on clutch operation?

- A) Clutch chatter during engagement — excessive free travel destabilizes the clutch disc
- B) No effect — clutch free travel has no impact on clutch engagement or disengagement
- C) The clutch may not fully disengage, making gear changes difficult and causing gear grinding
- D) Clutch slips under load — excessive free travel means the pressure plate remains applied too lightly

Q24. What voltage is standard for most North American heavy truck electrical systems?

- A) 24V
- B) 48V
- C) 6V
- D) 12V

Q25. A turbocharged diesel engine has an intercooler. The engine produces less power than specified and intake temperature is higher than normal. A pressure test of the intercooler and charge air piping shows the system holds pressure. What else should be checked?

- A) The intercooler is performing correctly — high intake temperature with no leaks means the turbocharger is the problem
- B) Check intercooler mounting — vibration loosens the intercooler and reduces its effectiveness
- C) Increase turbocharger boost pressure — the intercooler efficiency is being overcome by insufficient initial boost
- D) Perform a flow restriction test across the intercooler — it may hold pressure yet transfer heat poorly

Q26. A trailer is equipped with an anti-lock braking system (ABS). After ABS repairs, the technician notices the ABS warning lamp stays on at all times during driving. What is the likely cause?

- A) The lamp is supposed to stay on — it only goes out during ABS modulation
- B) An active fault code remains — not cleared after repair or newly introduced
- C) Trailer ABS lamps are driven by the tractor ECU — it needs a software update
- D) The ABS is in bypass mode — the lamp shows normal non-ABS braking

Q27. What is the purpose of the tractor protection valve on a tractor-trailer combination?

- A) To prevent the trailer brakes from applying during normal service braking
- B) To limit maximum air pressure to the trailer to prevent brake lockup
- C) To isolate the tractor air supply if the trailer breaks away
- D) To allow the driver to apply trailer brakes independently of tractor brakes

Q28. A diesel engine's EGR (Exhaust Gas Recirculation) valve is stuck in the open position. What symptoms would this cause at highway cruise speeds?

- A) Engine overheating — recirculated exhaust gases are very hot, increasing coolant temperature
- B) Turbocharger surge — excess flow from the EGR valve exceeds turbocharger capacity
- C) Rough running, loss of power, and black smoke as oxygen-depleted exhaust dilutes the intake charge
- D) No symptoms at cruise — EGR only operates at idle and low loads, not at highway speeds

Q29. A truck's alternator is producing 14.2V but the battery is not fully charging after long trips. What should be suspected?

- A) The alternator voltage is too high — it is overcharging the battery
- B) The voltage regulator is stuck in the low-voltage position
- C) High resistance in the charging circuit or a failing battery
- D) The battery needs distilled water — low electrolyte prevents charging

Q30. A newly installed driveshaft vibrates at a frequency approximately twice per revolution of the shaft. What type of vibration is this and what is the most likely cause?

- A) Third-order vibration from an incorrect number of balance weights — three weights cause three pulses per revolution
- B) Second-order vibration from improper U-joint phasing or excessive operating angle
- C) Random vibration from loose driveshaft companion flange bolts — not frequency-correlated
- D) First-order vibration from driveshaft imbalance, which occurs once per revolution

Q31. A truck's trailer light circuit has no tail lights on the trailer. The tractor tail lights work normally. The first check should be:

- A) Replace the trailer light module with a known-good unit
- B) Test the trailer harness from front to rear with a wiring tracer
- C) Check the 7-pin connector for corrosion or damaged pins
- D) Check the trailer ABS module — it controls all trailer lights

Q32. A truck driver reports a vibration that occurs only at highway speeds (above 90 km/h) and goes away below 70 km/h. The vibration is felt through the floor, not the steering wheel. What is the most likely cause?

- A) Engine misfire — engine misfires create vibration through the body at high RPM only
- B) Driveshaft imbalance or phasing error — felt through the floor within a specific speed range
- C) Front wheel imbalance — imbalance is always felt through the steering wheel
- D) Tire flat spot from hard braking — flat spots occur at any speed, not only above 90 km/h

Q33. A truck diesel engine cranks normally but will not start in cold weather. The intake air pre-heater (grid heater) has been verified as working. What should be checked NEXT?

- A) Check the ECM calibration level
- B) Check injector return (spill) volume
- C) Check fuel quality and filter
- D) Check the engine valve timing

Q34. A late-model diesel engine logs a fault for "NOx sensor downstream of SCR catalyst reading higher than expected." What is the MOST likely cause?

- A) DEF quality or concentration is out of spec, cutting SCR efficiency
- B) The DPF is loaded — soot backpressure is causing incomplete combustion
- C) The NOx sensor itself is failed high — replace sensor immediately
- D) The EGR valve is stuck open, routing excess exhaust into the intake

Q35. A carrier's safety manual cites NSC Standard 13 as the authority for its post-trip inspection policy. Which statement about post-trip inspections in the National Safety Code is correct?

- A) Post-trip inspection appears nowhere in the National Safety Code at all
- B) Standard 14 sets the post-trip interval used in the carrier safety rating
- C) Standard 13 requires a post-trip inspection report at the end of each day
- D) Standard 16 makes post-trip inspection a mandatory training competency

Q36. An air brake system governor is set at 125 PSI cut-out and 100 PSI cut-in. A technician resets the governor so that cut-out occurs at 175 PSI. What is the effect on system components?

- A) Pressure beyond what the dryer, valves, fittings and reservoirs are rated for
- B) No change — every component in an air brake system is rated to at least 175 PSI
- C) Stronger parking brake hold — the spring brakes clamp harder at higher pressure
- D) The compressor cycles more often because cut-out now sits further from cut-in

Q37. A truck clutch requires adjustment because the free travel has decreased to near zero. If ignored, what will result?

- A) Harsh and noisy gear engagement on every shift
- B) Increased effort needed at the clutch pedal
- C) Clutch slipping from release bearing contact
- D) Difficulty disengaging the clutch for shifting

Q38. A truck has 4 inches of free play at the steering wheel rim before the front wheels begin to move. What does the National Safety Code use as the limit for steering wheel free play?

- A) No limit at all — free play is not part of a safety inspection
- B) A flat 2 inches of free play at the rim on every commercial vehicle
- C) A flat 1 inch of free play at the rim on every commercial vehicle
- D) Limits that vary with wheel diameter and manual or power steering

Q39. An SCR (Selective Catalytic Reduction) system requires Diesel Exhaust Fluid (DEF). What happens if the DEF tank runs empty?

- A) The engine runs normally without any DEF
- B) Engine power is derated, eventually limited to idle
- C) The engine shuts off immediately
- D) A warning light comes on but operation continues indefinitely

Q40. A trailer spring brake chamber has failed on the air side and the spring brake on the right rear will not release. The chamber housing is intact. The immediate action before repair is:

- A) Fill the chamber with compressed air and seal it
- B) Disconnect the air line to prevent pressure from reaching that axle
- C) Cage the spring brake with its caging bolt
- D) Drive carefully to the shop with reduced speed

Q41. A truck's front axle service brakes are adjusting automatically, but during an inspection the left front is found to have significantly more push-rod travel than the right. What is the MOST likely cause?

- A) The left front linings are worn thinner than the right and cannot be taken up
- B) The left front S-cam was installed reversed and has to be turned back around
- C) The air line to the left front chamber is kinked, cutting its supply pressure
- D) The left front automatic slack adjuster is not maintaining push-rod travel

Q42. Under Canadian regulations, what is required when transporting dangerous goods (DG) by road?

- A) A dangerous goods endorsement added to the driver's licence
- B) Training, shipping documents, and vehicle placarding
- C) Shipping documents and placards, but training only for the shipper
- D) Placards on the vehicle and a training certificate, but no shipping document

Q43. A drive axle is fitted with a limited-slip differential. What does its oil need that a plain differential's oil does not?

- A) A detergent package to keep the clutches clean
- B) A higher viscosity oil than the plain axle uses
- C) A GL-4 rating instead of the usual GL-5 rating
- D) A friction modifier additive for the clutch pack

Q44. A driver discovers a major defect in the brakes during the daily trip inspection — the pushrod stroke is beyond maximum on three wheels. What must the driver do?

- A) Repair the brake adjustment himself before departure — drivers can perform minor brake adjustments to correct out-of-adjustment brakes
- B) Not operate the vehicle — report to the carrier, record the defect, and repair before operation
- C) Drive to the nearest repair facility — if the brakes still function, a major defect can be addressed after completing the current trip
- D) Record the defect at the end of the day — the report only covers defects found before the trip began

Q45. A tractor has had its trailer disconnect for one week. When reconnecting the trailer and attempting to release the trailer spring brakes via the trailer supply valve, the trailer spring brakes do not release despite the tractor system showing 120 PSI. The gladhand connections are confirmed secure. What should be checked first?

- A) Check the supply gladhand for an obstruction and the trailer emergency relay valve
- B) Manually cage all trailer spring brakes — they cannot be released remotely after long storage
- C) Replace the trailer spring brake chambers — long-term storage causes internal corrosion
- D) Increase tractor air pressure to 140 PSI — trailer spring brakes need higher pressure

Q46. A circuit protected by a 10-amp fuse keeps blowing the fuse. The CORRECT diagnostic approach is:

- A) Install a 20-amp fuse to stop the blowing
- B) Add a second 10-amp fuse in parallel
- C) Replace the fuse with a circuit breaker of any rating
- D) Find and repair the cause of the overcurrent

Q47. An engine oil cooler bypass valve opens when:

- A) Oil is cold and the pressure differential is high
- B) The oil pump output exceeds system pressure
- C) Oil temperature exceeds the maximum limit
- D) Coolant temperature exceeds 95°C

Q48. A heavy truck carries a heavy braided strap between the engine block and the frame rail. What is it for?

- A) It carries the alternator's output across to the frame-mounted loads
- B) It bonds the engine to earth so static cannot build up on the block
- C) It returns starter and accessory current to the battery negative
- D) It stops the engine mounts from passing vibration through to the cab

Q49. A loaded truck equipped with ABS is braking normally on dry, level pavement, and the ABS activates on every stop. This MOST likely indicates:

- A) The ABS is sensing impending wheel lockup
- B) Air pressure is too high
- C) The ABS module has failed
- D) ABS is working perfectly — this is normal

Q50. A truck has multiple electrical accessories that stop working. The fuses are all good. What should the technician check NEXT?

- A) Check individual component grounds only
- B) Replace the BCM (Body Control Module)
- C) Check for a blown fusible link
- D) Replace the alternator

Answer Key & Explanations

Q1. Answer: D

Quick-release valve: rapid local air exhaust from brake chambers on release, allowing spring return and reducing brake drag. Without a quick-release valve, air would have to travel all the way back through the treadle valve and exhaust at the driver's feet — taking too long and causing brake drag. The QR valve is located close to the brake chambers: when the pedal is released and line pressure drops, it immediately exhausts air locally at the chamber — rapid piston return and immediate brake release.

Q2. Answer: C

Low compression that improves on a wet test isolates the loss to the ring-to-bore seal. Oil poured into the cylinder temporarily fills the gap between the rings and the bore, so compression rises sharply. That narrows the fault to worn or broken rings or a worn, tapered or scored bore, but it does not by itself say which of the two — measuring bore taper and out-of-round after teardown separates them. If the wet test does NOT improve compression, the loss is past the rings: a valve not seating, a head gasket, or a cracked head or piston. Confirm with a cylinder leak-down test: apply compressed air with the piston at TDC and listen for air escaping at the intake (intake valve), the exhaust (exhaust valve), the crankcase breather (rings and bore), or the coolant (head gasket).

Q3. Answer: A

Excessive blow-by = combustion gases past piston rings. All engines have some blow-by — rings can't achieve a perfect seal. Worn or broken rings allow significantly more combustion gas into the crankcase, pressurizing it. Signs: oil pushed out seals and breathers, smoky breather outlet, rapid oil contamination. Measure with a blow-by flow meter to confirm.

Q4. Answer: B

Reject if the warning fails to activate, or fails to stay on, once pressure is lowered below 414 kPa (60 psi). National Safety Code Standard 11 Part B, Periodic Motor Vehicle Inspection, Section 3A — Air Brakes, lists this under low pressure warning. A visible warning is mandatory (lamp or wig-wag); an audible warning is optional but must still work when the manufacturer installed one. The same section carries a second, more serious tier: a warning device that is inoperative, or that fails to operate continuously with the ignition on and pressure below 380 kPa (55 psi), is a hazardous condition rather than a simple reject. Both figures are real and they do not conflict — 60 psi fails the inspection, 55 psi means the vehicle is too unsafe to be driven away.

Q5. Answer: A

Blue smoke = engine oil burning in the combustion chamber. Oil can enter combustion through: worn or broken piston rings (oil from crankcase), worn valve stem seals (oil from valve cover area enters intake), or failed turbocharger oil seal (oil from turbo enters intake or exhaust). Blue smoke at startup that clears = valve stem seals (oil pools during shutdown). Blue smoke that persists = rings or turbo seal. Test: remove air filter after a coast-down, check for oil in intake piping.

Q6. Answer: A

Relay valve: speeds up brake application at remote axles. Without a relay valve, air must travel the full length of the vehicle (foot valve to rear brakes). The relay valve uses a small pilot signal from the foot valve to open a local valve, delivering air from a nearby reservoir rather than the foot valve — much faster response.

Q7. Answer: C

Trip inspection report retention: at least 6 months. NSC Standard 13 requires carriers to retain the original copy of each vehicle inspection report and certification of repairs for at least 6 months from the date the report was prepared. The driver forwards the original to the carrier within 20 calendar days of completing it, must be in possession of the required inspection report while driving, and must produce it on demand of an inspector. Defects recorded on a report must be corrected before the next required inspection, or within the timeframe the jurisdiction of travel specifies, and a vehicle with a major defect must not be driven at all.

Q8. Answer: B

Dither: high-frequency low-amplitude signal superimposed on command to prevent solenoid stiction. Without dither, a proportional solenoid may stick at its current position due to static friction (stiction) between the plunger and bore. This causes hysteresis — the output at a given command differs depending on whether command is increasing or decreasing. Dither (typically 100–500 Hz) keeps the plunger in continuous slight motion, breaking static friction and ensuring the plunger position precisely tracks the command signal.

Q9. Answer: C

50/50 mix is standard for most conditions. This provides freeze protection to approximately -37°C and boilover protection to +106°C at atmospheric pressure, which a pressure cap raises further. Freeze protection comes from the glycol and the water interfering with each other, so it bottoms out near a 60/40 to 70/30 mix and then gets worse: straight concentrate actually protects far less than 50/50 (pure ethylene glycol freezes at about -13°C), and it also carries heat away less well than a water mix. Do not exceed 70% antifreeze.

Q10. Answer: D

Failed open, the valve stops isolating - a downstream leak can bleed the supply reservoir back through it. NSC Standard 11 identifies the one-way check valves as sitting between the supply (wet) tank and the service tanks; each service reservoir charges through its own valve, and the one-way action is what keeps a rupture in that circuit from emptying anything upstream. With the valve stuck open, a leak in that service circuit also drains the supply reservoir, and because the compressor is then feeding the leak, the other service circuit - still protected by its own check valve - keeps only its stored air and cannot be recharged. Failed closed is the opposite defect: that reservoir cannot charge at all, its low-air warning stays on and the vehicle is out of service. Test the valves by charging the system fully, then draining the supply reservoir: each service reservoir must hold its pressure.

Q11. Answer: D

Slip yoke: allows driveshaft length change as suspension cycles. When the rear axle moves up and down over bumps, the distance between the transmission output and the axle pinion changes. The slip yoke telescopes on the output shaft splines to accommodate this length change. Without a slip yoke, suspension movement would either stretch or compress the driveshaft, causing severe vibration or binding. The slip yoke splines must be lubricated to prevent wear and noise.

Q12. Answer: C

Oscilloscope: captures the waveform over time — reveals intermittent faults. A DMM reads average/RMS voltage and misses brief dropouts. An oscilloscope displays the actual PWM waveform — a dropout, missing pulse, or duty cycle change appears instantly on screen. Essential for intermittent electronic faults on PWM-controlled circuits.

Q13. Answer: A

A tire inspection covers four things - depth, condition, inflation and matching - and mixing a radial with a non-radial on one axle fails on the matching criteria. The tire criteria for a periodic inspection sit in National Safety Code Standard 11, Part B, Section 9 - Tires and Wheels, and they run across four items: tread depth, tread condition, tire sidewall and manufacturer markings, and tire inflation pressure. Matching belongs to the sidewall and markings item, under matching and application, which rejects when the nominal tire size difference on an axle is greater than 25 mm, when dual-mounted tire diameters differ by more than 13 mm, when wheel or rim size does not match tire size, when a required tire is missing, when a radial tire is mixed with a non-radial on an axle, or when a tire is labelled in a way that shows it is not intended for on-road use. Bias and radial tires on one axle are listed as a hazardous condition as well, so that truck is not fit to drive away either. The other three findings are not criteria: duals 8 mm apart sit inside the 13 mm limit, a retreaded tire is barred from an active steering axle rather than from a drive axle, and nothing in the section turns on who made the tire or how old it is - matching is judged by size, construction and diameter. Correct pressure and legal tread satisfy only two of the four things a tire inspection has to cover.

Q14. Answer: A

Coupling is not complete without a visual inspection — a tug test alone can pass on a false (high) couple. Ease forward with trailer brakes applied to feel resistance, then secure the vehicle and physically look: locking jaws fully closed around the kingpin shank (not on top of or ahead of it), no visible gap between the trailer upper coupler plate and the fifth wheel surface, and the release handle fully seated in the locked position. A kingpin riding on top of the fifth wheel plate or caught ahead of the jaws can hold during a tug test and then drop the trailer on the road.

Q15. Answer: A

Parallel batteries share one terminal voltage, so the bank is only as good as its worst battery. Wiring in parallel holds the system at 12 volts and adds cranking capacity, but it also ties every battery to the same voltage. A tired battery sitting beside a new one becomes a load on it: the new battery pushes current into the old one instead of into the starter, and it sits at a partial state of charge that shortens its own life. This is why heavy truck batteries are replaced as a matched set of the same type, rating and age, and why the starter feed and the ground are taken from opposite corners of the bank so current has an equal path through each battery. Series wiring is what raises voltage — parallel wiring never does.

Q16. Answer: C

Above 350 mm nominal, the fallback limit is 3.0 mm over the original diameter. National Safety Code Standard 11 Part B, Periodic Motor Vehicle Inspection, Section 3A — Air Brakes, rejects a drum whose measured diameter exceeds the limit indicated on the drum, the OEM standard, or an industry standard. Only when none of those is available does the standard supply its own figures: 2.3 mm over original diameter for a nominal drum size of 350 mm (14 in) or less, and 3.0 mm over original for a nominal size greater than 350 mm. A 420 mm drum falls in the larger group, so 3.0 mm applies. The order matters as much as the number — the marking on the drum comes first, and these figures are the last resort.

Q17. Answer: B

Clutch slipping: oil/grease contamination on friction surface — pressure plate cannot grip the disc. Oil from a leaking rear main seal or improperly lubricated parts contaminates the disc friction material. Oil-soaked friction material has greatly reduced coefficient of friction. Other causes: worn friction disc (too thin, past minimum thickness), weak or broken pressure plate springs, overheated pressure plate (springs lose tension from clutch abuse). Contamination: replace both disc and pressure plate — cleaning oil from friction material is ineffective.

Q18. Answer: A

One gear only, and only on the overrun: the clutching teeth for that gear are worn. In a non-synchronized heavy-duty box the sliding clutch is held on the gear by its clutching (dog) teeth. Repeated pop-out tapers those teeth, and once tapered they act like a ramp: the reversed driveline load of coasting drives the sliding clutch back off the gear, which is why the fault shows on deceleration and in that gear alone. A blocker ring cannot be the cause, because a non-synchronized main section has no synchronizers to wear. Loose mounts and a slipping clutch act on the whole drivetrain and would not single out one gear. Read the pattern first: a fault confined to one gear points at that gear pair, while a fault in every gear points at the shift rails, detents or linkage.

Q19. Answer: B

Maximum continuous operating angle: approximately 3 degrees. U-joints run at large angles generate torsional excitation and wear rapidly; higher angles are tolerated only at lower shaft speeds. Two companion rules: the operating angles at the two ends of the shaft must be equal within 1 degree, and every joint needs at least 1 degree of angle so its needle rollers keep rotating instead of brinelling the cross. Cancellation of the speed fluctuation requires BOTH conditions met at once - equal angles at the two joints AND correct phasing, meaning the in-board yokes of the shaft in line with each other. A shaft that is out of phase, or whose joint angles are unequal, leaves the second joint adding to the first joint's fluctuation instead of cancelling it. Check phasing and measure the angles as separate steps of a driveline inspection, and remember ride height changes the angles.

Q20. Answer: C

15% fuel consumption increase: systematic diagnostic required. Key suspects: leaking/dribbling injectors and injector balance rates (if one injector is fuel-trimming heavily), injection timing drift (worn pump timing), boost leak reducing combustion efficiency (turbo hose, CAC leak), increased parasitic load (seized compressor, alternator), or EGR valve stuck open (diluting air charge). Use a scan tool to check fuel balance, boost pressure, and injection timing.

Q21. Answer: B

Looseness in line with the ball shank beyond 3.0 mm, where the manufacturer publishes no figure, is the hazardous condition. National Safety Code Standard 11 Part B, Periodic Commercial Motor Vehicle Inspections, Section 4 — Steering, closes each item with a Hazardous Condition list that sits apart from the Reject If column. Under Item 1, Steering Control and Linkage, the hazardous entry for steering linkage reads: any ball and socket joint has looseness in line with the shank or neck of the ball greater than manufacturer specification, or when specification is not available, greater than 3.0 mm. Take the reading along the shank and not across it, using the tools and method the manufacturer's service instructions call for. The other three findings are Reject If entries in the same section, so the truck fails and no decal is issued, but the standard does not carry them onto a hazardous list. A split bellow or a missing clamp belongs to Item 1's bellow, clamp and boot entry. Less than 25 mm between the tire and the frame, fender or any other vehicle part at any point in the turn is Item 3's tire clearance entry. Rotations from centre to full left that differ from centre to full right by more than half a turn is Item 3's rotation and travel entry. Item 3's own hazardous list is short: steering that binds or jams, and lash past a second figure set above the one that rejects.

Q22. Answer: C

Three defective brakes out of ten is 30% — at or over the 20% threshold, so the vehicle goes out of service. The North American Standard Out-of-Service Criteria, administered by the Commercial Vehicle Safety Alliance and applied by Canadian enforcement at roadside, place a vehicle out of service when the number of defective brakes is equal to or greater than 20% of the service brakes on the vehicle or combination. A brake 6.4 mm (1/4 in) or more beyond its adjustment limit counts as one defective brake; a brake found exactly at the limit is not counted at all. The criteria's own chart shows that two defective brakes are enough to declare a ten-brake vehicle out of service, so three is well over. NSC Standard 11 notes that these roadside criteria are similar but not identical to its own shop-based Hazardous Conditions.

Q23. Answer: C

Excessive clutch free travel = insufficient pedal travel remaining to fully disengage clutch. Free travel is the pedal movement before the throwout bearing contacts the release fingers. After free travel is used up, the remaining pedal stroke releases the clutch. If free travel is too large, there isn't enough remaining pedal travel to fully disengage — the clutch drags (partially engaged) during shifting, causing gear grinding and premature synchronizer wear.

Q24. Answer: D

12V is standard for most North American trucks. European trucks commonly use 24V. Some newer hybrid systems use 48V. Two 12V batteries wired in parallel = 12V (more capacity). Two wired in series = 24V (used in European and some heavy equipment).

Q25. Answer: D

Intercooler can hold pressure but have poor heat transfer — bent fins, oil contamination, or internal blockage. A pressure leak test confirms no external air loss, but does not confirm thermal efficiency. If the intercooler core fins are bent (reducing airflow through the matrix), coated with oil from turbo leakage (oil acts as insulation), or internally scaled, heat transfer is reduced — outlet temperature remains high. Measure inlet and outlet air temperatures: large Δt = good. Also check ambient air restriction to the intercooler.

Q26. Answer: B

ABS warning lamp on during driving = active ABS fault code. Lamp sequence: illuminates at key-on for bulb check, then extinguishes after a short self-test if no faults. Continuous illumination during operation indicates an unresolved stored fault. After ABS repairs, fault codes must be cleared using a service tool or diagnostic software. If the lamp re-illuminates, a new fault was introduced during the repair (wiring disturbed, sensor reconnected incorrectly) or the existing fault was not actually repaired.

Q27. Answer: C

Tractor protection valve: isolates tractor if trailer breaks away. If the trailer supply line is severed or pressure drops below the control valve threshold (typically 45 PSI), the tractor protection valve closes, preventing tractor air from draining through the broken connection. This allows the tractor to maintain air pressure and braking capability. The trailer spring brakes also apply simultaneously.

Q28. Answer: C

EGR stuck open at cruise: rough running, loss of power, excessive exhaust smoke. At highway speeds, the engine needs maximum fresh air for full-load combustion. EGR gases are depleted of oxygen — introducing them at cruise reduces the effective air charge. The combustion is oxygen-starved, causing rich combustion (black smoke), loss of power, and rough running. The EGR is typically commanded closed at high load/high RPM by the ECM — a stuck-open mechanical valve overrides this. The engine may also misfire.

Q29. Answer: C

Correct alternator voltage but poor charging: check current flow and circuit resistance. 14.2V is within spec, but if the charging circuit has high resistance (corroded battery cable or charge wire, poor battery terminal connection), current flow to the battery is restricted despite correct voltage output. A battery with high internal resistance also won't accept charge efficiently. Test: measure charging current (clamp ammeter), measure voltage drop in the charging circuit, load test the battery.

Q30. Answer: B

Vibration twice per revolution is second order, and it comes from U-joint phasing or an excessive operating angle. A single U-joint operating at any angle creates velocity variation — the output shaft alternately speeds up and slows down twice per input revolution, which is second order. Two correctly phased U-joints cancel that variation. Incorrect phasing, meaning the yokes at the two ends are not in the same plane, makes the second variation add instead of cancel and produces a strong twice-per-revolution vibration. Excessive operating angle at either joint increases the magnitude of the variation as well, so a shaft can be correctly phased and still shake if the angles are large or unequal. Because the shaft is newly installed, the first thing to check is whether it went back together in the orientation it came apart in.

Q31. Answer: C

Trailer lights out: start at the 7-pin connector, and check the trailer circuit fuse. The 7-way connector is the most common cause of trailer lighting issues. Moisture, corrosion, and physical damage to pins cause high resistance or open circuits. Clean with electrical contact cleaner, inspect pins for spread contacts or corrosion. Also check the tractor-side fuse for the trailer circuit.

Q32. Answer: B

Speed-specific floor/seat vibration = likely driveshaft balance or phasing issue. Driveshaft vibration is proportional to rotational speed (square of speed for imbalance). At low speeds, even significant imbalance produces tolerable vibration. Above a threshold speed, vibration becomes severe. A phasing error — the yokes at each end of the shaft not aligned in the same plane — leaves the two joints' speed fluctuations adding instead of cancelling, which creates a second-order vibration. Front imbalance vibrates through the steering column/wheel. Rear driveshaft vibration transmits through the cab floor. Confirm with a driveshaft balance check and universal joint phasing inspection.

Q33. Answer: C

Cold no-start with a working intake pre-heater points at the fuel, not the air. Check the fuel grade against the temperature, the filter and water separator for wax, and the lines for gelling. Winter-grade or blended diesel has a lower cloud and pour point than summer grade; below the cloud point wax crystals form and pack the filter media, so the engine cranks, builds compression and gets heated air but never receives enough fuel. Confirm with a restriction reading or by warming and replacing the filter, then look for air drawn in on the suction side. Injector return volume, valve timing and ECM calibration are all real checks, but none of them explains a fault that appears only when the temperature drops.

Q34. Answer: A

DEF quality and concentration are the first thing to suspect when SCR conversion efficiency drops. Selective catalytic reduction uses diesel exhaust fluid, a 32.5 percent urea solution, to convert NO_x to nitrogen and water. Diluted, contaminated, or frozen-then-improperly-thawed fluid degrades that conversion, and the downstream NO_x sensor sees the result. The same fault family is triggered by low fluid level, a dosing injector fault, a fluid heater fault, and a degraded catalyst. A loaded particulate filter does not raise downstream NO_x, and an EGR valve stuck open lowers NO_x rather than raising it. Verify fluid quality and level before condemning the sensor or the catalyst.

Q35. Answer: D

Standard 13 never uses the term; Standard 16 does, and marks it mandatory. National Safety Code Standard 13 is the trip inspection standard, and the words post-trip and pre-trip do not appear anywhere in it. What it requires is a Schedule 1 inspection every 24 hours, an inspection report that records the fact where no defects are found, and a duty on the driver to monitor the vehicle while driving, record any defect detected on the inspection report and report it to the carrier before the next required inspection — immediately if the defect is a major one, in which case the vehicle is not to be driven. Post-trip appears instead in Standard 16, Entry Level Training, where competency 7.2.11, 'Conducts regular enroute and post-trip vehicle inspections', is marked M for mandatory; items marked R for recommended sit in the same table, so the marking is unambiguous. Standard 15, the facility audit standard, has a section headed Pre/Post Trip Inspections and defines a Pre/Post Trip Inspection violation as an audited finding against the carrier. Standard 14, Safety Rating, contains no occurrence of the term at all. The carrier's policy is real enough; Standard 13 is simply the wrong authority to hang it on.

Q36. Answer: A

A 175 PSI cut-out drives the supply side past what the dryer, valves, fittings and reservoirs are built to hold.

The Official Air Brake Handbook sets the ceiling plainly: actual cut-out pressure must never be higher than 1000 kPa (145 psi), and cut-out normally sits only 138 to 173 kPa (20 to 25 psi) above cut-in. Safety valves normally open at 1035 kPa (150 psi), so a 175 PSI setting leaves the reservoir safety valve venting continuously — and a safety valve that is venting air means too much pressure is building in the system and needs immediate repair. Seals, hoses, the dryer purge valve and fittings all work above their design pressure, and the compressor runs loaded far longer than intended. The extra pressure buys nothing at the brakes: application pressure is metered by the treadle and relay valves, and the spring brakes are held off by air but applied by their springs, so parking hold does not change. Reset the governor to the OEM cut-in and cut-out figures with a calibrated gauge.

Q37. Answer: C

Zero free travel leaves the release bearing touching the pressure plate, which bleeds off clamping force. Even slight continuous contact takes load off the friction facings, and the clutch slips under torque. It also spins the release bearing all day under load, which is what kills the bearing. Free travel — the pedal movement before the release bearing reaches the release fingers — exists so that the pressure plate is fully clamped whenever the pedal is up. It is set to the figure in the manufacturer's service information and measured at the pedal pad, and it closes up on its own as the friction facings wear, which is why it is a scheduled check rather than a one-time adjustment. The release bearing to clutch brake clearance is a separate adjustment with its own figure and is set in the same procedure.

Q38. Answer: D

Free play has no single limit — it changes with steering wheel diameter and with manual versus power steering. National Safety Code Standard 11 Part B, Section 4 — Steering, sets the Canadian figures, and each province adopts them in its own inspection regulation. Measure at the rim with the engine running for power assist, turning the wheel each way until the front wheels just begin to move. The vehicle is rejected when lash exceeds 75 mm for a power-steering wheel of 500 mm or less, or 87 mm for one over 500 mm; manual steering is rejected past 87 mm and 100 mm. A second, higher set of figures in the same item marks a hazardous condition. So 4 inches (about 102 mm) is past every power-steering reject limit, but no truck can be judged until the wheel is measured and the steering type noted.

Q39. Answer: B

DEF empty = progressive engine derate. SCR requires DEF (urea solution) to reduce NOx emissions. When DEF runs out, the system cannot meet emission requirements. The ECM first warns, then progressively reduces power (derate), eventually limiting the engine to idle. This prevents NOx emission violations and encourages immediate refill — adding DEF restores normal operation.

Q40. Answer: C

The caging bolt mechanically compresses the power spring and holds the brake released. Each spring brake chamber has a caging bolt for exactly this purpose — compressing the large power spring so the brake releases when air cannot release it. Chock the wheels first, and never cage a chamber whose housing is cracked, corroded or damaged: the power spring can be ejected. A caged spring brake has NO spring braking ability at that wheel. Under the periodic inspection standard, a spring brake caged by a caging bolt or made inoperative by other mechanical means is a rejection item, and an inoperative brake is a major defect on the trip inspection schedule. The unit must not be operated in service — tow it, or move it only as the jurisdiction permits to reach repair.

Q41. Answer: D

One side over-stroking while the other holds spec points at that side's automatic slack adjuster. An automatic slack adjuster in working order takes up lining-to-drum clearance a little at a time as the linings wear, holding applied push-rod travel roughly constant across the life of the linings - which is why linings worn thinner on one side are not, by themselves, an explanation: a functioning adjuster absorbs the difference. Measure applied stroke with 620-690 kPa (90-100 psi) in the tanks, engine off, spring brakes released and the service brakes fully applied. A seized, worn or incorrectly installed adjuster stops taking up wear, and stroke creeps out on that wheel alone. A kinked supply line slows application rather than lengthening a full-application stroke measured at working pressure, and a reversed S-cam would never have worked from the day it was assembled. NSC Standard 11 notes that where stroke on a self-adjusting brake exceeds the limit, a manual adjustment will not correct the problem - repair the cause, then re-measure.

Q42. Answer: B

The Transportation of Dangerous Goods Act, 1992 and its Regulations set three duties that travel together. Anyone who handles, offers for transport or transports dangerous goods must be adequately trained and hold a valid training certificate, or do the work in the presence and under the direct supervision of someone who is — so training does not stop at the shipper. A shipping document must accompany the shipment, and the vehicle must display the correct safety marks for the class and quantity being carried. An Emergency Response Assistance Plan is required for certain higher-risk goods. There is no dangerous goods endorsement on a Canadian driver's licence; the training certificate is the credential.

Q43. Answer: D

A friction modifier. A limited-slip differential adds a clutch pack or a cone clutch between the side gears and the differential case, so that when one wheel starts to spin some torque is still carried across to the wheel that still has grip. Those clutches are meant to slip a little and grip a little every time the axle goes round a corner, because the two wheels are travelling different distances and the differential has to let them. Plain hypoid gear oil makes them grab and release in steps rather than slipping smoothly, and the axle chatters and shudders through slow tight turns while the clutch faces glaze. The friction modifier is an additive that narrows the gap between static and sliding friction in the clutch pack so the slip is smooth and quiet. Some gear oils are sold already modified for limited-slip service; otherwise the modifier is added separately in the quantity the axle maker states. The other three answers describe choices made for other reasons: viscosity and the gear-oil rating are chosen for the ring and pinion and do not change because the differential is limited-slip, and a detergent package belongs to engine oil, which has combustion by-products to carry, not to axle oil.

Q44. Answer: B

Major defect found at the daily trip inspection: the vehicle must NOT be operated until it is repaired and certified. National Safety Code Standard 13 splits every schedule into a defect column and a major defect column, and the two carry different duties. Schedule 1 lists pushrod stroke of any brake exceeding the adjustment limit as a major defect, so one wheel over stroke already qualifies and three wheels is far past it. Standard 13 then states that no motor carrier shall permit a person and no person shall drive a commercial vehicle on a highway when a major defect is present. The driver records the defect on the inspection report, does not operate the vehicle, and reports the condition to the carrier immediately. The carrier has the defect repaired and certified before the vehicle returns to service. A driver may not correct this by adjusting the brakes at the roadside, and there is no allowance for finishing the current trip first.

Q45. Answer: A

Trailer spring brakes not releasing: check the supply line and trailer emergency (relay) valve first. When trailer air lines are disconnected, the trailer emergency valve detects loss of supply and applies the trailer spring brakes (fail-safe). On reconnection some trailer emergency valves do not reset if supply pressure rises too slowly or if the valve is sticking — verify it is receiving supply and passing it to the spring brake chambers. Also verify the couplers are on the right ports: in Canada the supply (emergency) gladhand is red and the service gladhand is blue, and the supply line must be connected and sealing before the trailer can charge. Crossed lines put supply air into the service line, so the trailer tanks never fill and the spring brakes cannot release. Damaged gladhand seals leak away the supply pressure.

Q46. Answer: D

Never increase fuse rating as a fix. The fuse protects the wiring, not the component. A blown fuse means the circuit is drawing more current than the wire can safely carry. Increasing the rating allows wiring to overheat and potentially start a fire. Diagnose why the circuit draws excess current — short circuit, overloaded component, damaged wiring.

Q47. Answer: A

Oil cooler bypass valve: opens on cold thick oil. Cold engine oil has high viscosity and creates a high pressure differential across the oil cooler. The bypass valve opens to route oil around the cooler, ensuring adequate lubrication flow during cold starts before oil warms and thins. Once warm, the valve closes and oil flows through the cooler normally.

Q48. Answer: C

It is the return path for every current that flows through the engine, and above all for the starter's. A starter draws several hundred amps, and its case is bolted to the engine, so the engine block is one end of the starter circuit. That current has to get back to the battery negative, and it will use whatever conducting path it can find. The braided strap exists to give it a short, low-resistance path of the right cross-section. Where the strap is missing, loose or corroded, the current goes looking for another route: through throttle and shift cables, through air lines and their fittings, through the coolant in a heater hose, through a fuel line, or through the shield of an engine control harness. The symptoms are correspondingly odd and scattered — slow cranking, lights that dim when an unrelated load is switched, burned cable jackets, pitted module ground pins, and datalink faults that come and go. The strap is braided rather than solid because the engine moves on its mounts and a solid conductor would work-harden and crack, and both ends must be cleaned to bright metal. It is a current path, not an anti-static device and not a vibration control, and the alternator has its own output cable and does not use it.

Q49. Answer: A

On dry pavement, ABS should only intervene during hard or emergency braking. The system cycles only when it calculates that a wheel is decelerating far faster than the others and is about to lock, so cycling on every ordinary stop is a symptom to be traced. Either a wheel really is heading for lockup — an over-adjusted, grabbing or dragging brake, poor brake balance, or a badly worn or mismatched tire — or the ABS is being told so falsely by a defective wheel speed sensor, an excessive sensor air gap, or a damaged or contaminated tone (exciter) ring. Check brake adjustment and balance, then read the wheel speed signals with a scan tool. On ice, packed snow or wet pavement the same cycling is normal system operation, and a failed module would disable ABS and light the warning lamp rather than cycle it.

Q50. Answer: C

Multiple circuits dead with good fuses = upstream fault. A blown fusible link or tripped main circuit breaker affects all circuits downstream. Fusible links are located near the battery and protect main wiring harnesses — they look like wire but contain a sacrificial element.

Scored under 70%?

The real exam draws from a much wider pool. The **310T Truck & Transport Mechanic Complete Question Bank** gives you every question on our platform with full explanations and key-concept summaries — built to close exactly the gaps this mock exposed.

Get the 310T Truck & Transport Mechanic Complete Question Bank →

Or keep practising free online at redsealquiz.ca/310t.html

© 2026 · redsealquiz.ca/310t.html · For personal study use only. Independent, unofficial study resource — not affiliated with or endorsed by the CCDA, the Red Seal Program, or any apprenticeship authority. See "About This Material".